An Audit of Acute poisoning at QECH

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Background



Objective and Methods



Results



Discussion

Background

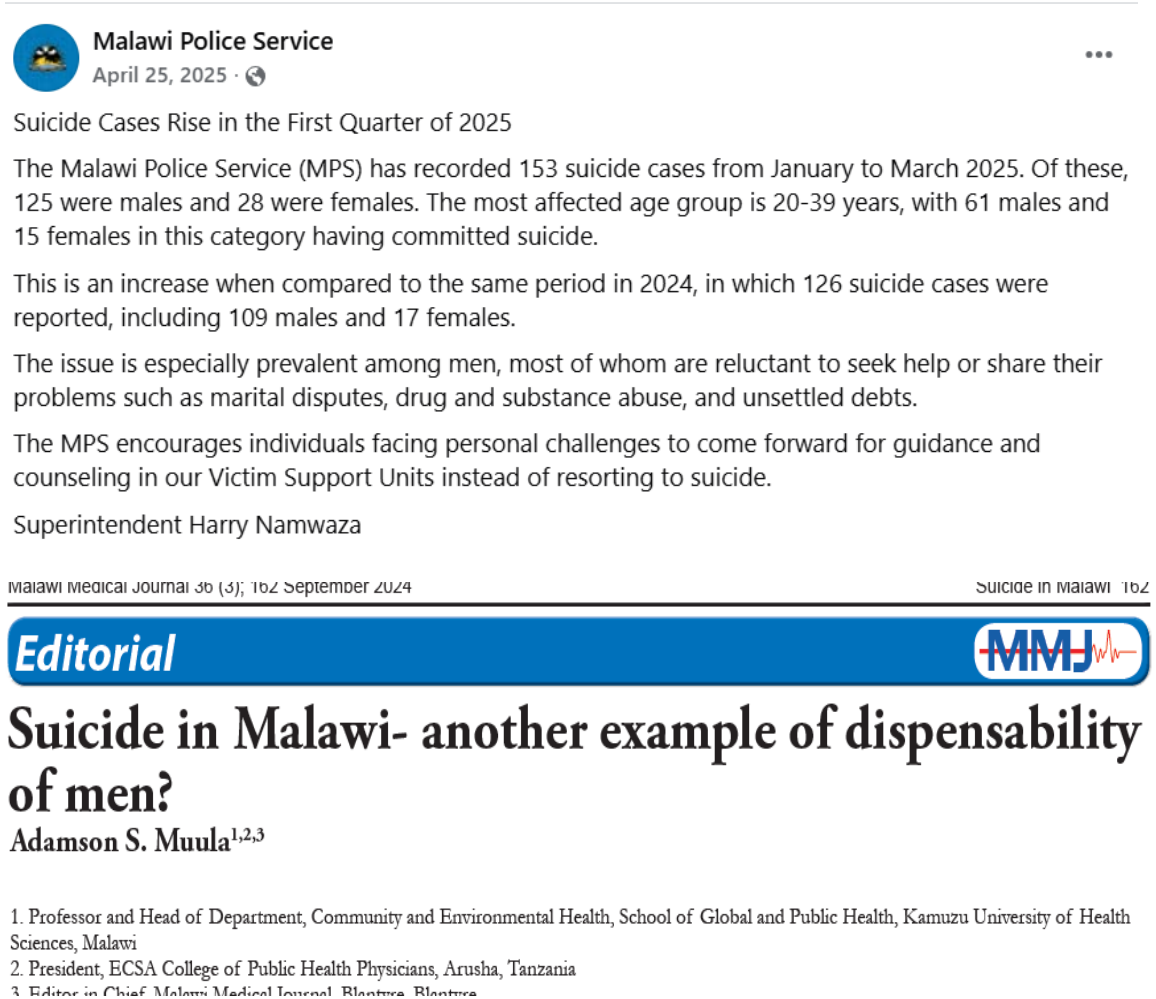
- Acute poisoning is defined as exposure to poisoining within 24 hrs whether intentional or non-intentional which causes adverse effect
- World wide about 193,000 deaths were attributed to be due to poisoning(WHO 2019)
- In Africa, about 134/788 (1.7%) of presentation to the ED are acute poisoning; 87% of which have been documented to be intentional poisoning



Background

- In Malawi, suicide rates have been on the rise in Malawi according to the Malawi Police service …126 in Q1 2024→ 153 cases in Q1 2025, predominantly affecting males

- Common reasons for suicides include substance abuse; marital disputes or debt



Malawi Police Service
April 25, 2025 ·

Suicide Cases Rise in the First Quarter of 2025

The Malawi Police Service (MPS) has recorded 153 suicide cases from January to March 2025. Of these, 125 were males and 28 were females. The most affected age group is 20-39 years, with 61 males and 15 females in this category having committed suicide.

This is an increase when compared to the same period in 2024, in which 126 suicide cases were reported, including 109 males and 17 females.

The issue is especially prevalent among men, most of whom are reluctant to seek help or share their problems such as marital disputes, drug and substance abuse, and unsettled debts.

The MPS encourages individuals facing personal challenges to come forward for guidance and counseling in our Victim Support Units instead of resorting to suicide.

Superintendent Harry Namwaza

Malawi Medical Journal 36 (3); 162 September 2024 Suicide in Malawi 162

Editorial MMJ

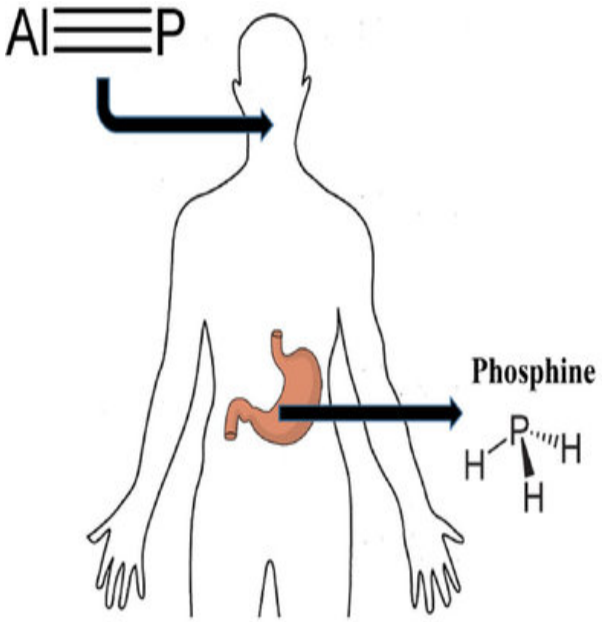
Suicide in Malawi- another example of dispensability of men?

Adamson S. Muula[1,2,3]

1. Professor and Head of Department, Community and Environmental Health, School of Global and Public Health, Kamuzu University of Health Sciences, Malawi
2. President, ECSA College of Public Health Physicians, Arusha, Tanzania
3. Editor in Chief, Malawi Medical Journal, Blantyre, Blantyre

Background continued

- Aluminium phosphide “Chimbulu” is a pesticide if ingested by people that has a high fatality rate of 77-100%.



Mechanisms of toxicity:

Induction of cellular hypoxia and free-radicals-mediated injury
Inhibition of cytochrome c oxidase and vital cellular enzymes
Production of reactive hydroxyl radicals

Clinical manifestations:

Early symptoms:

Circulatory collapse (the major lethal consequence)
Other features may include dizziness, fatigue, tightness in the chest, headache, nausea, vomiting, diarrhea, ataxia, numbness, paresthesia, tremor, muscle weakness, diplopia and jaundice

Late symptoms:

Hepatorenal toxicities

Malawi News Agency
April 8 ·

PCB warns against illegal use of aluminium phosphide

Lilongwe, April 8, Mana: Pesticides Control Board (PCB) has warned the public against illegal sale, distribution and use of aluminium phosphide, a highly toxic pesticide restricted for professional use.

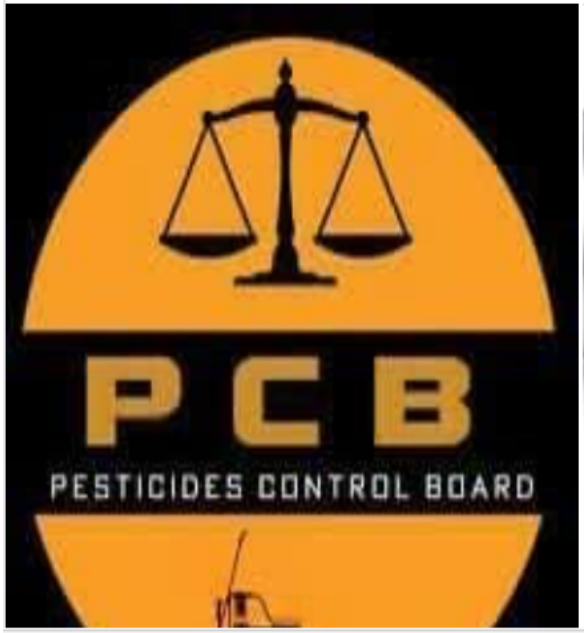
In a public notice released Wednesday, PCB says the chemical must only be handled by trained and certified commercial pesticide applicators and sold by licensed dealers.

The Board expressed concern that some agro-dealers and grain traders are selling and using the pesticide without following legal procedures, putting lives at risk.

“Aluminium phosphide has a high mortality rate if mishandled and is only permitted for indoor fumigation in sealed storage facilities,” reads the statement in part.

Those found contravening the law risk up to two years imprisonment or a fine of K5 million.
The Board has since urged the public to report illegal sale or use of the pesticide to relevant authorities.

By Roy Howa





Rationale of audit

- Increasingly high number of patients presenting with aluminium phosphide poisoning with different clinical presentation from other organophosphate poisoning

- Need to describe the type of patients who present with aluminium phosphide poisoning and compare their clinical presentation to other types of acute poisoning patients that present to Queen Elizabeth Central Hospital

Objectives

- We aimed to describe the patients who presented with acute poisoning at QECH
- We also compared clinical presentations and outcomes among those who ingested “chimbulu” vs those who did not.
- We also evaluated the factors associated with mortality among patients presenting with acute poisoning at QECH.

Methods



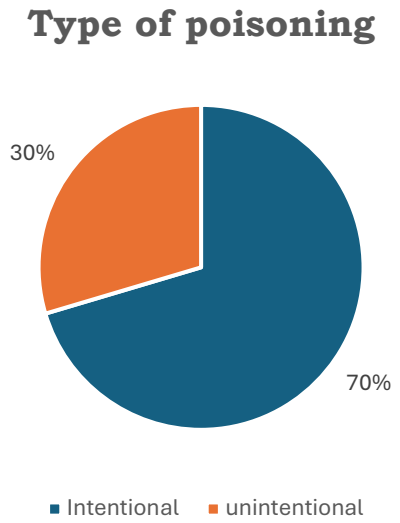
- **Study design**
  - Audit carried at QECH AETC.
  - Patients who had a diagnosis of poisoning in MAHIS from using ICD 11 codes between 01 May 2025 and 31 Mar 2026 were include in the analysis
- Data collection
  - MAHIS officer shared an Excel file with all patients who had diagnosis of acute poisoning → individual patient information including clinical notes were reviewed to confirm if they met criteria → additional variables were added to excel file based on clinical notes
  - Definitions
    - Hypoglycemia -glucose <3.0 mmol
    - Hypotension- relative systolic <100 mmHg
    - Low GCS- if ever recorded to have GCS <15 or lost consciousness
    - Urban or rural classification based on address
- Data analysis
  - Data was analysed using STATA 15.

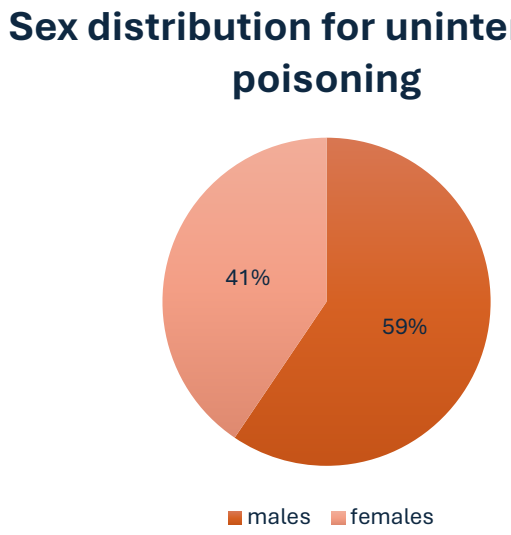
Results

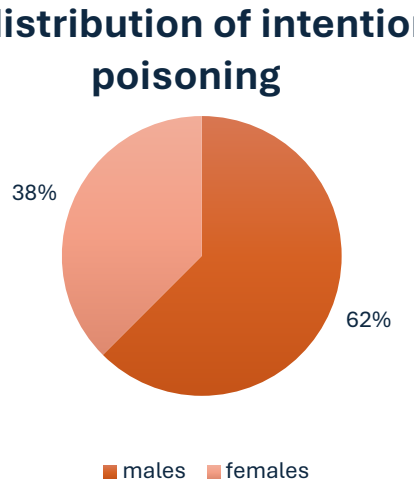
- A total of 132 records review; 125 confirmed cases of poisoning; 7 were other diagnoses

- Of 125 patients, 62 % males (n=77) while 38% (n=48)females

- Most common age group was 20-30 years (44%)







Variable	n (%) Total (N=125)	Unintentional (n=37)	Intentional (n=88)	
			Other poisoning (n=60)	Chimbulu (n=28)
Age category				
<20	26 (20.8%)	9 (24.3%)	11 (18.3%)	6 (21.4%)
20–30	55 (44.0%)	12 (32.4%)	34 (56.7%)	9 (32.1%)
30–40	23 (18.4%)	6 (16.2%)	10 (16.7%)	7 (25.0%)
40–50	12 (9.6%)	5 (13.5%)	4 (6.7%)	3 (10.7%)
>50	7 (5.6%)	5 (13.5%)	0 (0.0%)	2 (7.1%)
Median age (IQR)	25.0 (20.8–33.7)	27.4 (20.3–40.0)	22.8 (20.8–28.8)	28.0 (21.2–37.2)
Sex				
Male	77 (61.6%)	22 (59.5%)	37 (61.7%)	18 (64.3%)
Female	48 (38.4%)	15 (40.5%)	23 (38.3%)	10 (35.7%)
Residence				
Urban	81 (64.8%)	23 (62.2%)	39 (65.0%)	19 (67.9%)
Rural	44 (35.2%)	14 (37.8%)	21 (35.0%)	9 (32.1%)

Table 1: Demographic characteristics of patients

Most of the patients were young males from urban region

Table 2: Type of poisoning agents

Poison category	**n (%) Total (N=125)**	**Unintentional (n=37)**	**Intentional (n=88)**
Food poisoning	22 (17.6%)	22 (59.5%)	0 (0.0%)
Unknown pesticide/insecticide	19 (15.2%)	6 (16.2%)	13 (14.7%)
Unknown	20 (16.0%)	0 (0.0%)	20 (22.7%)
Rat tablet /Chimbulu	28 (22.4%)	0 (0.0%)	28 (31.8%)
Organophosphates/Pesticides	12 (9.6%)	3 (8.1%)	9 (10.2%)
Pharmaceuticals/Drugs	11 (8.8%)	0 (0.0%)	11 (12.5%)
Household chemicals/other	9 (7.2%)	2 (5.4%)	7 (7.9%)
Herbal/Traditional	4 (3.2%)	4 (10.8%)	0 (0.0%)

Most of ones with unintentional poisoning were due to food poisoning; while as intentional poisoning was due to pesticide use

Reasons for intentional poisoning

Of those documented 13 cases were due to Romantic disputes; 7 family disputes; 5 personal problems; 1 financial dispute

"hx of unresponsiveness after taking a tablet called chimubulu around 12pm……

Guardian reports that three days ago notified that he would better take poison himself due to being swindled money by his girlfriend"

"34 y/o male presenting to the AETC for confusion, vomiting, and suicidal ideations. Reports taking pesticides yesterday after noon after drinking alcohol because he wanted to end his life. His wife has recently left him because of his gambling addiction"

"38yo male
history of ingesting rat poison around 6am after being heart broken by finding his wife with another man"

"22 YO female who presented 2 hours after taking an known poisonous tablet because she wanted to commit suicide after a disagreement with her husband"

"He is a student at MUBAS in first yr, first semester.
He says he failed one course out of the six courses he currently has , this made him disappointed and he intentionally bought rat poison in form of a big pill which he crushed and ingested all its powder"

Table 3. Clinical Presentation of patients with acute poisoning

Those with unintential poisoning presented with vomiting, abdominal and AVPU score was Alert

Those with suspected chimbulu poisoning Had low GCS; hypotension; Hypoglycemia; vomiting And only half were Alert based On AVPU score

Variable	n (%) Total (N=125)	unintentional (n=37)	Intentional poisoning (n=88)	
			Other poisoning (n=60)	Chimbulu (n=28)
Vomiting	58 (46.4%)	19 (51.4%)	26 (43.3%)	13 (46.4%)
Salivation/Secretions	12 (9.6%)	1 (2.7%)	6 (10.0%)	5 (17.9%)
Abdominal Pain	39 (31.2%)	18 (48.6%)	18 (30.0%)	3 (10.7%)
Cardiac arrhythmia	11 (8.8%)	0 (0.0%)	5 (8.3%)	6 (21.4%)
Diarrhoea	19 (15.2%)	9 (24.3%)	7 (11.7%)	3 (10.7%)
Low GCS	45 (36.0%)	7 (18.9%)	22 (36.7%)	16 (57.1%)
Hypotension	27 (21.6%)	4 (10.8%)	4 (6.7%)	19 (67.9%)
Hypoglycaemia	13 (10.4%)	0 (0.0%)	1 (1.7%)	12 (42.9%)
AVPU Score				
Alert	98 (78.4%)	35 (94.6%)	49 (81.7%)	14 (50.0%)
Verbal	8 (6.4%)	1 (2.7%)	1 (1.7%)	6 (21.4%)
Pain	6 (4.8%)	0 (0.0%)	3 (5.0%)	3 (10.7%)
Unresponsive	5 (4.0%)	0 (0.0%)	3 (5.0%)	2 (7.1%)

Majority (49%) of those with unintentional poisoning were discharge

Majority (38%)of those with other unintentional poisoning were admitted

Majority (50%) of those with chimbulu died

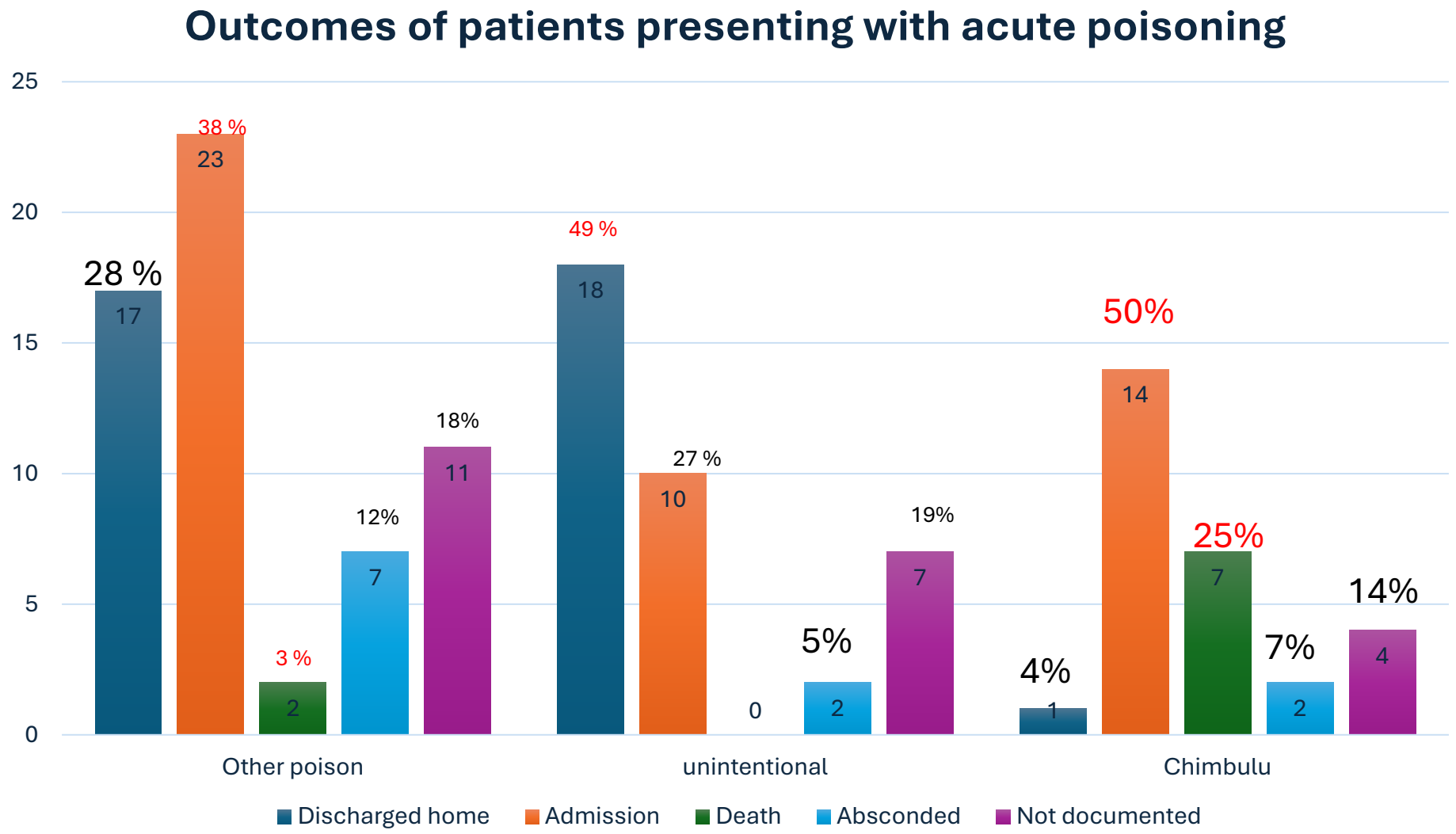


Table 4 Factors Associated with Mortality (Univariate Analysis)

Hypotension; vomiting and low GCS were significant predictors of mortality

Variable	Death/Admission (n=56)	Discharge/Abscond (n=47)	p-value
Hypotension	18 (34.6%)	4 (8.9%)	0.003
Tachycardia/arrhythmia	8 (14.8%)	2 (4.4%)	0.106
Age (median, IQR)	25.0 (20.3–33.7)	24.0 (20.3–32.2)	-
Male sex	36 (64.3%)	28 (59.6%)	0.686
Vomiting	28 (87.5%)	19 (59.4%)	0.022
Diarrhoea	11 (37.9%)	6 (27.3%)	0.552
Abdominal pain	14 (66.7%)	23 (88.5%)	0.086
Urban residence	-	-	-
Low GCS	29 (58.0%)	7 (15.9%)	<0.001

Table 5. Multivariable Logistic Regression for Mortality

Chimbulu poisoning, low GCS and vomiting were strong predictors of mortality

Variable	Adjusted OR	95% CI	p-value
Chimbulu poisoning	4.2	0.83 – 21.23	0.083
Hypotensive	1.79	0.48 – 6.65	0.381
Low GCS	4.49	1.55 – 13.03	0.006
Vomiting	2.42	0.12 – 0.54	0.061

Discussion

- The rates of unintentional poisoning are similar to those described elsewhere
  - 64% in Ethiopia; 87% Martinique France
  - However other settings describe females- 55% as being the ones to ingest poisonings; other female to male ratio of 1:1.06; others 57% male
  - Age group of ingestion is same 20-30 years in Ethiopian but a bit older >40 years of age in western settings
  - Common used poison were pesticides while as other settings – pharmaceutical drug overdose
- Clinical presentation was different from other organophosphate poisoning
  - Commonly presented with hypotension;low GCS ; hypoglycemia in line with pathophysiology- release of cytotoxic phosphine gas causing circulatory collapse
  - Compare to other OPP which causes cholinergic symptoms after inhibition of acetylcholinesterase

Discussion cont..

- Mortality due to aluminium phosphide was high
  - Other settings have noted mortality rates 50-100%
  - Dose dependent and affected by time to presentation
  - No specific antidote noted
    - Other studies recommend an anti-oxidant to fight the oxidative stress- ?? Magnesium sulphate- not conclusive
    - Most treatment provided was supportive

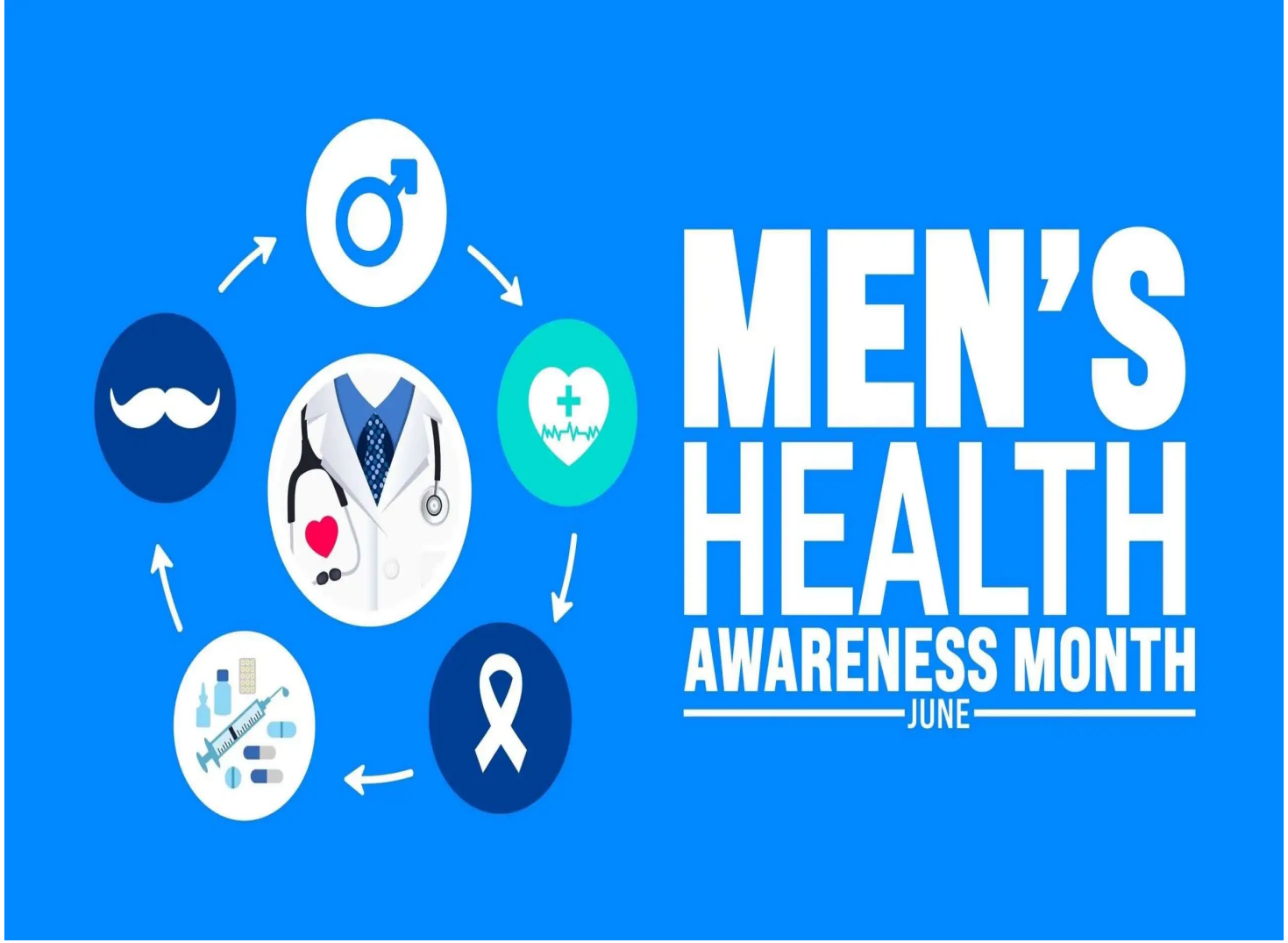
Table 2: Cases of patient survival following aluminum phosphide poisoning

Author	Case report
Dueñas *et al.*[10]	Trimetazidine was successfully used for reversal of cardiovascular manifestations of phosphine poisoning
Shadnia *et al.*[11]	A 28-year-old man survived following rapid treatment by coconut oil which prevented absorption of ALP 6 h postingestion
Siddaiah *et al.*[12]	Intraaortic balloon pump was used for toxic myocarditis from ALP poisoning in a young female until the effects of ALP positively resolved
Mehrpour *et al.*[13]	Digoxin was used for the management of cardiogenic shock in an 18-year-old female with acute ALP poisoning until the effects of ALP resolved. The patient was discharged 10 days after admission, with full recovery
Hassanian-Moghaddam *et al.*[14]	A 28-year-old ALP-poisoned male was put on venoarterial extracorporeal membrane oxygenation, following which blood lactate level decreased and cardiac function improved. He was discharged in stable condition
Marashi *et al.*[15]	A 40-year-old male patient with suicidal ingestion of one tablet of ALP in cardiogenic shock was started on hydroxyethyl starch. At the end of therapy, the patient was stable and discharged with normal clinical and paraclinical examinations

Limitations

- No standard form- based on clinical notes
  - Misclassification of poison ..rat tablet vs chimbulu
  - Time to presentation, dose amount of poison , antidote given missing
  - Information on lab test or other investigations done variable
- Outcomes may not have been updated

Key take away points from Audience……


MEN'S
HEALTH
AWARENESS MONTH
JUNE

References

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